Lecture 10

Sequencing Concepts

Nomenclature

- **Forward**: The "forward" strand was designated when the data was standardized. There is nothing special about the forward strand that would distinguish it from the reverse.
- **Reverse**: The reverse complement of the **forward** strand.
- **Sense**: The same orientation as the original single-stranded DNA.
- **Antisense**: The reverse complement of the original DNA.

Sequencing randomly sheared DNA

Starts with multiple copies of DNA:

```
==========================================
==========================================
==========================================
```

Each copy sheared into fragments of various size

```
=====   ====   ===== ====== ===========
===   ==========   ==== ===========   ==
=======   ========   ===== ===========
```

From double strands to single strands

Each double stranded DNA fragment:

```
====================
```

Is split into strands `+` and `-`:

```
(5')++++++forward++++++(3')

(3')------reverse------(5')
```

Single strands have a directionality indicated by the words `5'` (five prime) and `3'` (three prime)

Most processes operate from `5' -> 3'`

Sequencing also operates from 5' to 3

Different ends will be sequenced into "reads" on the forward and reverse strands of the same fragment:

```
---read -->
++++++++++++++++++
```

On the reverse strand:

```
------------------
        <---read--
```

The instrument can only sequence fragments within a certain size range. Sizes that are too long or too short won't work at all.

Single end sequencing

Each read corresponds to different, randomly, single strand DNA fragments. The read may represent the forward or the reverse strand of the fragment

```
---read -->
++++++++++++++++++
```

Then from a **different** fragment we may get:

```
------------------
        <---read--
```

All data goes into the same file. One file per sample.

Paired-end sequencing

Read pairs will be generated from a randomly selected, single strand DNA fragments.

```
---read -->
++++++++++++++++++
```

In a second step the **same** DNA fragment is reverse complemented and sequenced again:

```
------------------
        <---read--
```

We typically get two files. The **first** and **second** pair. We must keep them synchronized.

Read pair orientation

A shorthand notation (head to head, innie):

```
-------> <-------
```

First in pair does not mean that the first read comes from the forward strand. It is the "first" to be sequenced.

The *second in pair* is the reverse complement of first but we don't know beforehand which strand it is on.

This pairing is the most common Illumina technology.

Benefits of paired-end sequencing

The method sequences both ends of the **same** fragment.

- More reliable when locating regions in the genome. Both ends need to match.
- It can bridge over unknown regions. Better assembly.
- The same fragment sequenced twice. Error correction may be possible.

Recommended practice for any method where we study variation or assemble sequences.

Downsides of paired-end sequencing

The method sequences both ends of the **same** fragment.

- We measure the same thing twice. We may have redundant data.
- It has a higher cost (~25% more expensive).
- It has twice the runtime.

Paired-end is NOT recommended practice for approaches where the fragments are short or you want to count unevenly covered data: ChIP-Seq, RNA-Seq.

Strand specific sequencing

The instrument keeps track of which strand the original DNA came from and removes the other strand from sequencing.

The first read of the fragment indicates the orientation.

```
------>
```

The read corresponds to the original orientation of the fragment

Strand specific caveats

The strand-specific process is library preparation dependent.

Some protocols (like the most popular Illumina Strand Specific TruSeq) will match on the **antisense** rather than **sense**.

There is a reverse transcription process along the way. So you get the results backwards 😕

The first read matches on the reverse complement. It is easy to recognize, but adds to the complexity.

Strand-specific paired-ends

Visually this can be confusing.

We will see some examples later.

You have to distinguish between first in pair and second in the pair.

Here the second in the pair will be in the correct (**sense**) orientation.

Other paired-end methods

Other methods are also in use. Here is a slang:

1. Innie (FR, forward-reverse):

```
-------->  <--------
```

2. Outie (RF, reverse-forward):

```
<--------    -------->
```

3. Tandem (FF, RR):

```
-------->    --------->
<--------    <---------
```

What to do for other methods?

The `FR` (innie) `------> <----` method is dominant, and most methods will work with these.

For other orientations, you may either:

1. Pick a tool that recognizes the orientation.

2. Reverse complement one or more read pairs to bring them into the correct orientation.

Sequencing coverage (depth)

The base formula:

```
COVERAGE = TOTAL_SEQUENCED_BASES / GENOME_SIZE
```

When using a sequencing instrument:

```
COVERAGE = SUM_OF_ALL_READ_LENGHTS / GENOME_SIZE
```

Expressed as `1x` or `10x.`

Indicates, on average, how many times each base was measured (covered by a read).

Some people call this **sequencing depth**

Sequencing coverage in practice

Coverage example:

```
---------
    ------------
        -------------
              ----

1112223332223332111
```

The bottom line is the *local coverage* at each base

The coverage is the average of those numbers is `2`.
So we call that `2x`.

Round to the nearest integer

Coverage for constant read lengths

The formula becomes:

```
C = N * L / G
```

Where :

```
C = Coverage
N = Number of Reads
L = Read length
G = Target Genome Size
```

Questions everyone asks

What coverage do I need?

How much data should I collect?

How many samples should I sequence?

Use the formula

The formula (approximate values are ok):

```
C = N * L / G
```

What is the coverage of 100 million, 100 bp long reads over the human genome?

```
C = 100,000,000 x 100 / 3*10^9 = 3.3 = 3x
```

What is the coverage of ten thousand, 100 bp long reads over the Ebola genome?

```
C = 10,000 * 100 / 18000  = 55.5 = 55x
```

More complicated questions

Realistic scenarios may be more complicated. For example:

1. An instrument produces **1 million** reads per lane.
2. The read length can be set from **50** to **250**.
3. Shorter reads cost a lot less.

Typical Question

I want to sequence **5** samples of a genome of size **3 million** bases. What should the read length be to get **10x** coverage for each sample?

Applying the formula

Start with:

```
C = N * L / G
```

Identify what you know:

```
N = 1,000,000
G = 3,000,000 x 5 samples
C = 10
```

Rearrange to solve for `L`:

```
L = C * G / N = 150
```

What if the coverage is not uniform?

The formula works only when the fragmentation is uniform. Each fragment has the same chance of appearing.

Functional assays measure variable abundances. This formula does not apply anymore.

In those cases, we read what other scientists have done and how well it worked out for them.

Use those as guidelines on what coverage to pick.

Probability of a base not sequenced

Lander/Waterman model (random fragments)

Probability of a base not being sequenced for a coverage C

```
P = exp(-C)
```

Examples:

```
C = 5, P = exp(-5) = 0.007 = 0.7%
```

Genome size = **250 million** --> **15 million** bases not sequenced!

Realistic coverages

Theoretical coverage predictions are not quite right

The empirical observation is that usually need to raise the required coverage at least 5-10 fold.

What part of the genome is coverable to begin with? Some regions do not show up at all. Why?

Terms people us: “accessible”, “mappable”, “effective” genome sizes

Effective genome sizes

The **effective genome size** is a scaling factor (usually less than 1).

Tries to account for unknown bases `N`. By that measure, the effective size of chrom22 is 80%.

Tries to account for repetitive regions that a given method may not be able to analyze.